Magnetic Force and Torque on Current-Carrying Wires (completed)

Goal.

- See where the force on a current-carrying wire, $F = IlB\sin\alpha$, comes from — it's just $q\vec{v}\times\vec{B}$ summed over every moving charge in the wire.
- Use the right-hand rule both ways: given I and $\vec{B}$, find $\vec{F}$; and given the $\vec{F}$ you need, find the current direction that produces it.
- Work a full numerical example end to end: find the current that lets a wire "float" against gravity in a magnetic field (Example 29.13).
- See why a current *loop* in a field feels a net torque even though it feels no net force, and derive $\tau = mB\sin\theta$.
- Work a full numerical example applying $\tau = mB\sin\theta$ to a multi-turn coil.

1. Where the force on a wire comes from

A current is moving charge. A short segment of wire, length Δx, carrying charge Δq with drift velocity $\vec{v}_\mathrm{d}$, feels the same magnetic force as any moving charge: $F = \Delta q\, v_\mathrm{d} B$. The charge crosses the segment in a time $\Delta t = \Delta x / v_\mathrm{d}$, so $\Delta q = I\,\Delta t$ (in terms of I and Δt). Substituting:

$$F = \Delta q\, v_\mathrm{d} B = \frac{\Delta q}{\Delta t}(v_\mathrm{d}\Delta t)B = I\,\Delta x\, B$$

Adding up every such segment over the full length l of wire (when the wire is perpendicular to $\vec{B}$) gives $F = IlB$. More generally, for a wire at angle α to the field:

$\vec{F}_\mathrm{wire} = I\vec{l}\times\vec{B} = (IlB\sin\alpha,$ **direction by right-hand rule**$)$

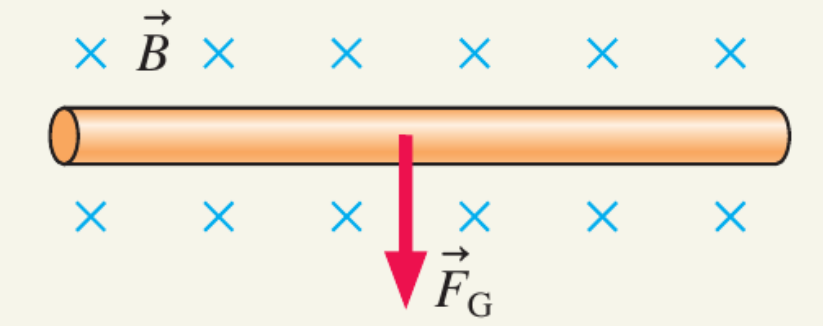
Check. The box above should look just like the moving-charge force law $\vec{F} = q\vec{v}\times\vec{B}$, with two substitutions. What plays the role of q, and what plays the role of $\vec{v}$?

The combination $I\vec{l}$ plays the role of $q\vec{v}$ — current times length stands in for "how much charge is moving, how fast," just as it did when this same force law was first built up from the individual charge carriers.

NOTE — Right-hand rule for a current-carrying wire: point your right thumb in the direction of the current (parallel to $\vec{l}$), your index finger in the direction of $\vec{B}$. Your middle finger then points along $\vec{F}$.

2. Worked example: magnetic levitation (Example 29.13)

A uniform field $B = 0.10$ T is horizontal, parallel to the floor. A straight segment of 1.0-mm-diameter copper wire, also parallel to the floor, is perpendicular to the field. What current through the wire, and in which direction, will allow the wire to "float" in the magnetic field?



Label $\vec{B}$'s direction, and sketch in the current direction and $\vec{F}_B$ once you've worked out which way they go.

MODEL. The wire floats if the magnetic force on it points upward and has magnitude mg, exactly balancing the downward gravitational force.

SOLVE — set the forces equal. With the wire perpendicular to $\vec{B}$:

$$F = IlB = mg$$

The weight comes from the copper's density $\rho = 8920$ kg/m^3 and the wire's volume. Write the mass as $m = \rho V$, with V the volume of a cylinder of radius r and length l:

$$IlB = \rho(\pi r^2 l)g$$

Check. One factor appears on both sides of that equation and cancels. Which one — and why should the current needed to float the wire *not* depend on how long the wire is?

The length l cancels. A longer wire weighs proportionally more ($mg \propto l$), but it also feels proportionally more magnetic force for the same current ($F \propto l$) — the two effects grow together, so the current that balances them is the same no matter how long the segment is.

Solve for the current:

$I = \frac{\rho\pi r^2 g}{B}$

Plug in $r = 0.00050$ m, $g = 9.80$ m/s^2, and the given B and ρ:

$$I = \frac{(8920)\pi(0.00050)^2(9.80)}{0.10} = 0.69 \text{ A}$$

Direction. With $\vec{B}$ pointing into the page (as drawn), the right-hand rule says the current must flow left to right to make $\vec{F}_B$ point up.

Check. The book notes this idea is only useful once you can get current into and out of the floating segment. If the lead-in wires run straight down to the segment (parallel to $\vec{B}$), why do *they* feel no force while the segment does?

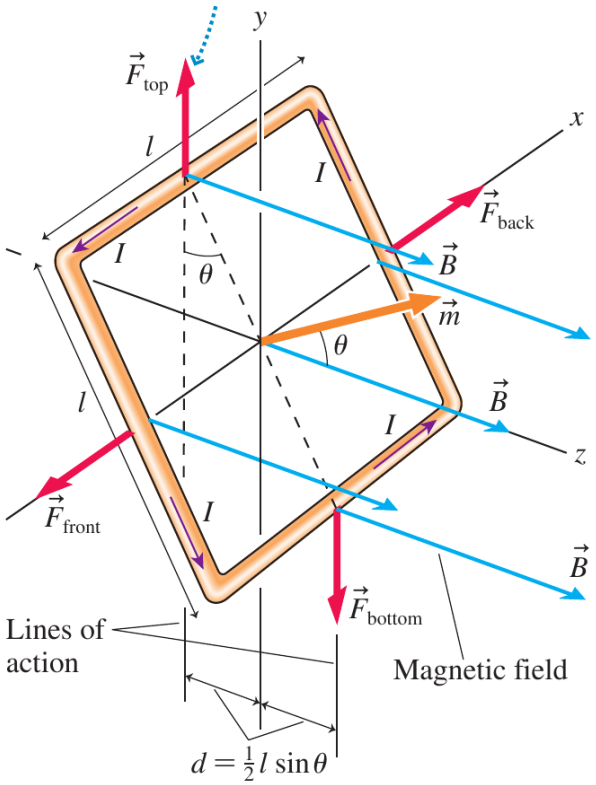
$\vec{F} = I\vec{l} \times \vec{B}$ vanishes whenever $\vec{l}$ and $\vec{B}$ are parallel (or antiparallel) — $\sin\alpha = 0$ there. The lead-in wires run along $\vec{B}$'s direction, so they feel no magnetic force at all; only the horizontal segment, which is perpendicular to $\vec{B}$, does.

3. Torque on a current loop

Now put a whole square loop (side l, current I) in a uniform field $\vec{B}$, with the loop's normal — its magnetic moment $\vec{m}$ — at angle θ to $\vec{B}$. The field pushes on all four sides, but two of those forces, $\vec{F}_{\text{front}}$ and $\vec{F}_{\text{back}}$, are equal and opposite *along the same line* and simply cancel. The other two, $\vec{F}_{\text{top}}$ and $\vec{F}_{\text{bottom}}$, are also equal and opposite — so the *net force* on the loop is still zero — but they act along different lines, separated by a moment arm d. A force pair like that doesn't translate the loop; it rotates it.

FIGURE 29.48 A uniform magnetic field exerts a torque on a current loop.

$\vec{F}_{\text{top}}$ and $\vec{F}_{\text{bottom}}$ exert a torque that rotates the loop about the x-axis.



Mark which pair of forces cancels outright, and which pair is the one producing the torque.

Each of $\vec{F}_{\text{top}}$, $\vec{F}_{\text{bottom}}$ has magnitude $F = IlB$ (Section 1's result, with $\theta = 90°$ between the wire and $\vec{B}$), and the moment arm for each is $d = \frac{1}{2} l \sin\theta$. Torque is force times moment arm, and there are two force/arm pairs acting together:

$$\tau = 2Fd = 2(IlB)\left(\tfrac{1}{2} l \sin\theta\right) = Il^2 B \sin\theta$$

The combination $Il^2 = IA$ (area $A = l^2$ for this square loop) is the loop's magnetic moment magnitude m (or $m = NIA$ for N turns instead of just one). This result turns out to hold for a loop of *any* shape:

$\tau = mB\sin\theta$

In vector form, since this is one more example of a cross product:

$\vec{\tau} = \vec{m} \times \vec{B}$

Check. At what angle θ is this torque zero? At what angle is it maximum? A compass needle is a magnetic dipole — use your answer to explain why a compass needle always ends up pointing along $\vec{B}$, not across it.

Zero at $\theta = 0°$ or $180°$ ($\vec{m}$ aligned or anti-aligned with $\vec{B}$); maximum at $\theta = 90°$ ($\vec{m} \perp \vec{B}$). The torque always acts to twist $\vec{m}$ toward alignment, so $\theta = 90°$ — pointing *across* the field — is exactly where the twisting force is strongest, not an equilibrium. The needle only stops turning once it reaches $\theta = 0°$, aligned with $\vec{B}$.

4. Worked example: torque on a coil

A flat coil has $N = 40$ turns, each a square $l = 3.0$ cm on a side. The coil carries a current $I = 1.5$ A and sits in a uniform field $B = 0.50$ T, with its magnetic moment at $\theta = 35°$ to the field (as in Figure 29.48). What is the magnitude of the torque on the coil?

MODEL. The coil is a magnetic dipole with moment $m = NIA$; the torque follows directly from $\tau = mB\sin\theta$.

SOLVE. First find the area of one turn and the dipole moment:

$$A = l^2 = (0.030\ \text{m})^2 = 9.0 \times 10^{-4}\ \text{m}^2$$

$$m = NIA = (40)(1.5\ \text{A})(9.0 \times 10^{-4}\ \text{m}^2) = 0.054\ \text{A·m}^2$$

Now find the torque:

$\tau = mB\sin\theta = (0.054)(0.50)\sin 35°$

Plug in the numbers:

$\tau = 0.027 \times 0.574 = 1.5 \times 10^{-2}\ \text{N·m}$

Check. If you doubled the number of turns N — same loop size l, same current I — would the torque go up, down, or stay the same? What does that suggest about why real motors and galvanometers wind *many turns* of wire around the same coil, rather than using a single loop?

$m = NIA$, so doubling N (with l and I unchanged) doubles m, and therefore doubles $\tau = mB\sin\theta$ too. Winding many turns around the *same* coil is a cheap way to multiply the torque for a given current and field — which is exactly why real motors and galvanometers use coils of many turns rather than a single loop of wire.